

My visualisation explores the impact of Australian bushfires, focusing on why bushfire severity cannot be understood through burned area alone. The target audience is the average Australian viewer who may know the 2019–20 Black Summer as a major disaster, but may not understand how physical fire scale, timing, economic exposure and wildlife vulnerability interact. The main purpose is explanatory: to guide viewers through a multi-layered story showing that fire impact depends not only on how much land burns, but also where it burns, what assets are exposed, and which ecosystems or species are already vulnerable.

The project combines several public datasets. Long-term burned-area patterns are based on Australian fire extent and forest fire datasets, including DCCEE/ERIN fire extent data and ABARES/Fires in Australia's Forests 2016–21 attributes. Satellite hotspot timing uses NASA FIRMS MODIS active fire detections, cleaned and aggregated by month, region and fire season. Economic impact is based on the Insurance Council of Australia Historical Normalised Catastrophe Data, using normalised insured losses and claims information. Wildlife impact uses WWF Australia's Black Summer animal impact estimate, while threatened species analysis uses bushfire-related species and EPBC status data. I processed the raw files into smaller derived CSVs for web performance, including monthly hotspot summaries, state-season burned-cell metrics, planned/unplanned burn shares, tenure-category-burn pathways, and event-level insurance/burned-area matches. Where data could not be honestly joined, such as state-level ICA loss for Black Summer, I avoided fabricating values and used event-level matching instead. Caveats are included where measures are proxies, such as burned raster cells rather than exact hectares.

The design follows Munzner's What/Why/How approach. The "what" includes spatial data, temporal data, categorical attributes and quantitative measures such as burned area, hotspot counts, insured loss, claims and species counts. The "why" is mainly to compare, summarise, locate patterns, identify outliers and explain relationships, helping viewers distinguish physical scale from social, financial and ecological severity. The "how" uses varied idioms matched to these tasks. A streamgraph shows thirty years of burned-area composition over time, while a state shift chart compares Black Summer against a pre-event baseline. A geographic hotspot map and bivariate burn map support spatial interpretation. A calendar heatmap reveals seasonal timing and the difference between national and southeast fire patterns. Small multiples show state-level fire profiles, and a Sankey/alluvial view connects forest tenure, forest category and burn outcome. Economic charts include a timeline, bubble chart, domestic-commercial claim composition, and an

event profile comparison of burned area versus insured loss. Wildlife charts use a waffle/pictorial matrix and threatened-species bar chart to separate animal-count scale from conservation vulnerability.

Interactive features such as hover tooltips, filters, toggles, highlighting and zooming are used only where they improve interpretation. The page is structured as a scroll-based story with section introductions, annotations, source labels and insight boxes, so the audience can move from national fire scale, to geography and seasonality, to economic exposure, and finally to ecological consequences. This structure supports presentation rather than open-ended analysis, making the visualisation accessible while still showing complex relationships across multiple datasets.